# Prabhavit Gupta

Solution Architect | LangGraph, Agentic AI Systems, MCP & Enterprise Architectures

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## Executive Summary


- **LangGraph & Multi-Agent Solution Architecture:** Software and solution architect with 3 years commercial experience designing and deploying enterprise AI agent platforms and autonomous agentic workflows. Specialised in LangGraph state machines, multi-agent collaboration, persistent checkpointing, dynamic tool calling, and human-in-the-loop approval architectures.
- **Enterprise Integration, MCP & Security Governance:** Deep hands-on experience bridging agentic AI with enterprise systems using Model Context Protocol (MCP Server & Client), REST/gRPC APIs, and vector/relational databases. Proven track record enforcing strict AI governance, prompt injection defense, credential isolation, and audit logging across critical workflows.
- **Observability, Evaluation & Cloud Engineering:** Proven expertise establishing end-to-end evaluation and monitoring pipelines (RAGAS, Langfuse, OpenTelemetry, LangSmith) to benchmark agent task accuracy, latency, and token cost. Adept at architecting scalable microservices on AWS/Azure/GCP with Docker, Redis, and PostgreSQL/pgvector.


## Technical Skills & Architecture Capabilities

- **Agentic AI & Orchestration:** LangGraph, LangChain, Multi-Agent Swarms, Tool Loops, ReAct Reasoning, Dynamic State Persistence, Checkpointing, Human-in-the-Loop Approval Gates, Context Pruning.
- **Enterprise Protocols & Integration:** Model Context Protocol (MCP Server & Client SDKs), REST APIs, WebSockets, gRPC, PostgreSQL (pgvector), Redis (Caching & Pub/Sub), Enterprise SaaS Integrations.
- **AI Security & Governance:** Prompt Injection Mitigation, Guardrails (NeMo / Llama-Guard), Least-Privilege Agent Tooling, Sensitive Data Masking, Audit Tracing, Deterministic vs. Probabilistic Boundaries.
- **Observability, Evals & LLMOps:** Langfuse Tracing, LangSmith, OpenTelemetry, RAGAS Benchmarking, Regression Evaluation Harnesses, Token Budgeting, Latency Profiling, Automated CI/CD.
- **Backend & Cloud Platforms:** Python (AsyncIO, FastAPI, PyTest, Pydantic), TypeScript, Docker, Kubernetes, AWS, Azure, GCP, Microservices Architecture.
- **Architecture Leadership:** Solution Design Documents (SDD), Architecture Decision Records (ADRs), Technical Stakeholder Alignment, Client Workshops, Engineering Mentorship.

## Professional Experience

**AI Solution Architect & Engineering Consultant** • *Independent Consultancy (VAT Registered)*
2024 – Present • Venice, IT (Remote Europe)

- Spearheaded architecture, technical design, and implementation of production agentic AI solutions and LangGraph workflows for European client organizations.
- Designed multi-agent architectures utilizing LangGraph state graphs with conditional branching, persistent memory checkpoints, and human approval interrupts for high-stakes business decisions.
- Developed and standardized Model Context Protocol (MCP) server architectures, enabling AI agents to securely query internal enterprise relational databases and external analytics APIs with strict role-based access.
- Established enterprise AI security boundaries: implemented input/output guardrails, sanitized model tool-call arguments, and enforced auditable action trails to mitigate prompt injection and data leakage.
- Designed comprehensive evaluation harnesses using RAGAS and automated regression suites, setting quantitative release gates for task completion rates, hallucination control, and token efficiency.
- Integrated distributed observability with Langfuse and OpenTelemetry, delivering real-time dashboards for tracing multi-turn agent paths, token expenditures, and system latency.
- Authored technical architecture blueprints, conducted client design reviews, and guided engineering teams through containerized production rollouts using Docker and cloud infrastructure.

**Full-Stack AI Software Engineer** • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Architected and deployed asynchronous microservices in Python (FastAPI) and TypeScript to support automated data extraction, parsing, and semantic classification pipelines.
- Implemented enterprise retrieval-augmented generation (RAG) platforms using PostgreSQL with pgvector and Redis, optimizing retrieval latency and embedding cache hit rates.
- Built robust test automation suites with PyTest and CI/CD pipelines via GitHub Actions, establishing reliable automated quality checks across distributed backend services.
- Collaborated with cross-functional business and engineering leaders to translate complex domain specifications into

scalable, production-ready system designs.

## Key Architectural Projects

- **Enterprise Multi-Agent Platform (LangGraph + MCP + PostgreSQL):** Multi-agent collaborative system coordinating specialized domain agents with shared state persistence, MCP tool integration, human-in-the-loop review nodes, and Langfuse distributed tracing.
- **Automated Financial Intelligence Pipeline (FastAPI + pgvector + RAG):** High-throughput enterprise document processing and semantic search architecture with Pydantic structured output validation, token budgeting, and automated RAGAS eval gates.

## Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  *Rigorous foundation in formal logic, statistical reasoning, decision theory, and complex systems analysis.*
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to work across the EU without visa sponsorship; European VAT (Partita IVA) ready for freelance contracts; available immediately for 40 hours/week.